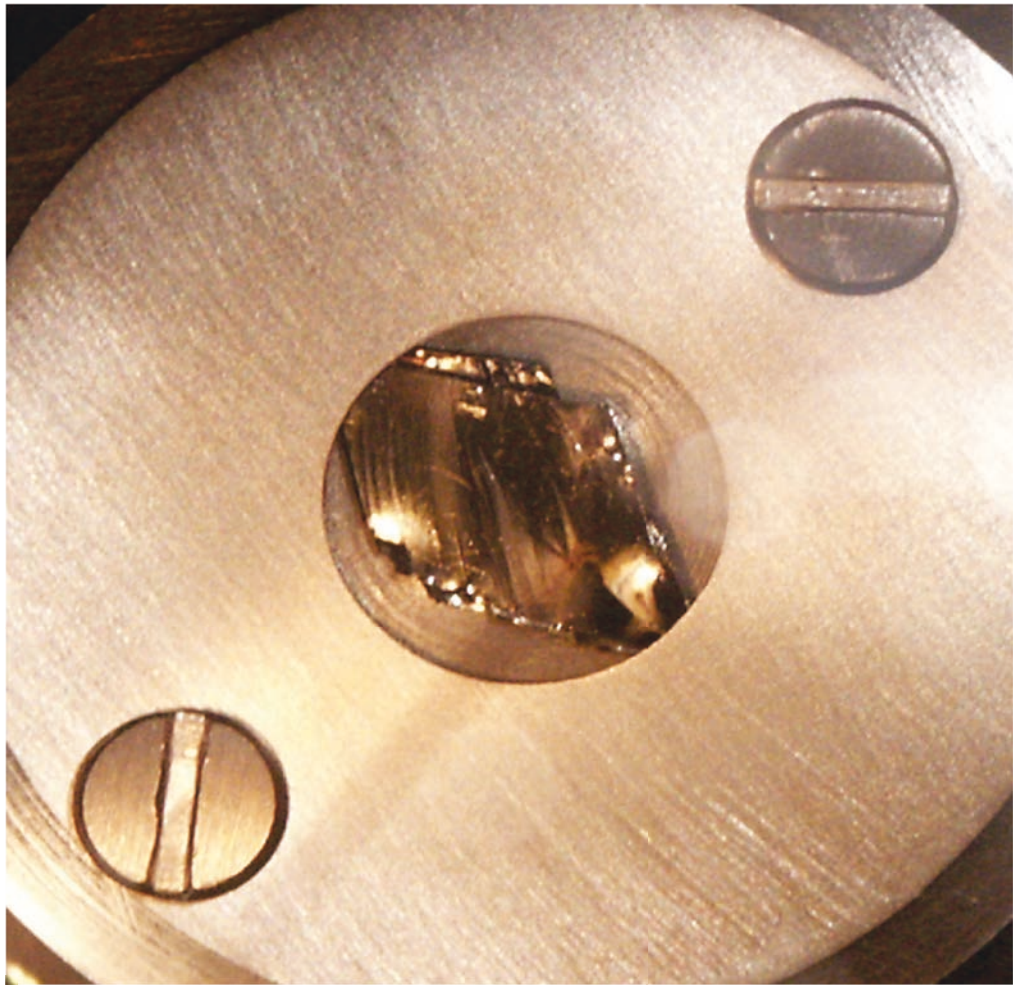
ARPES of Single Crystal Uranium (001)



Angle Resolved Photoemission Spectroscopy (ARPES) of Single Crystal Uranium (001)

Presented by:

Cyril P. Opeil, S.J.

In collaboration with:

R. K. Schulze, R. C. Albers, B. Mihaila, K. B. Blagoev, M. E. Manley, J. C. Lashley, J. L. Smith of the Los Alamos National Laboratory

and

P. B. Littlewood of the Cavendish Laboratory, University of Cambridge

Funding provided by the:

Department of Energy

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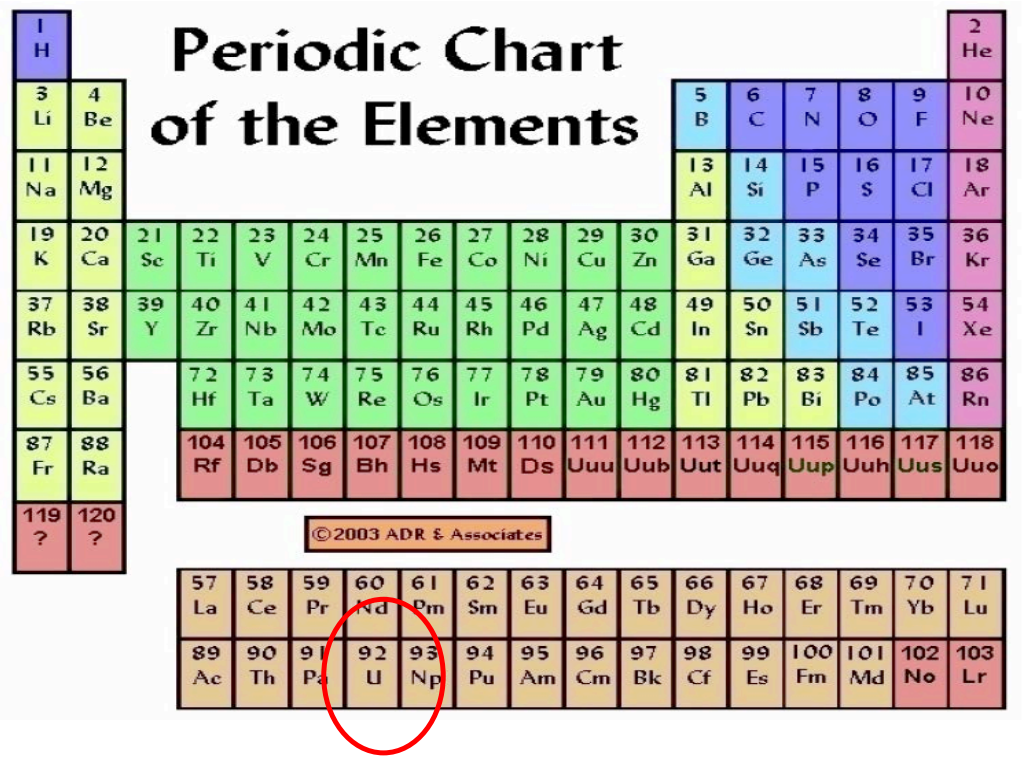
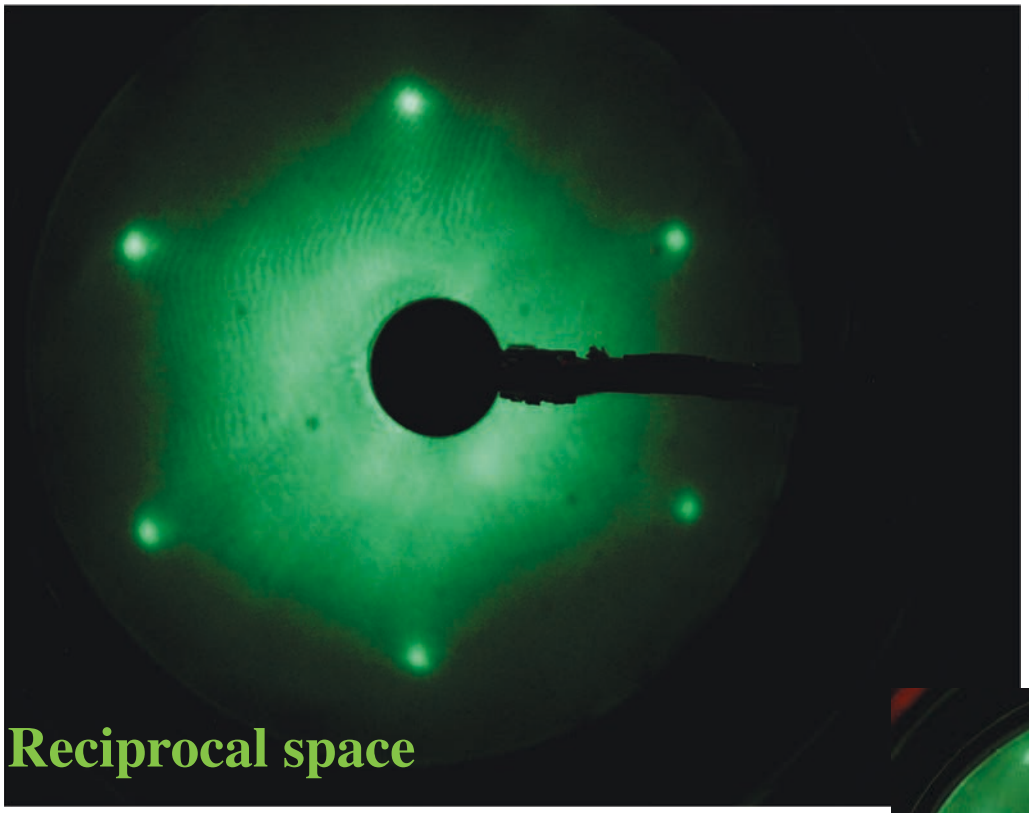
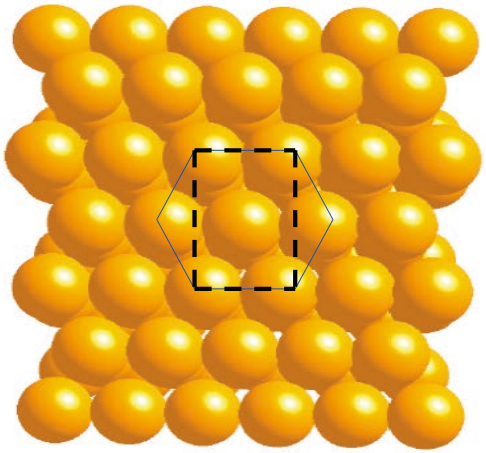
Why Study Uranium?



- Actinides possess itinerant to localized "f" electron behavior
- U(001) sample quality not previously available
- First comprehensive band map of an actinide now possible
- Actinide band map calculation -- improved accuracy

LEED pattern of U(001) at T = 273 K



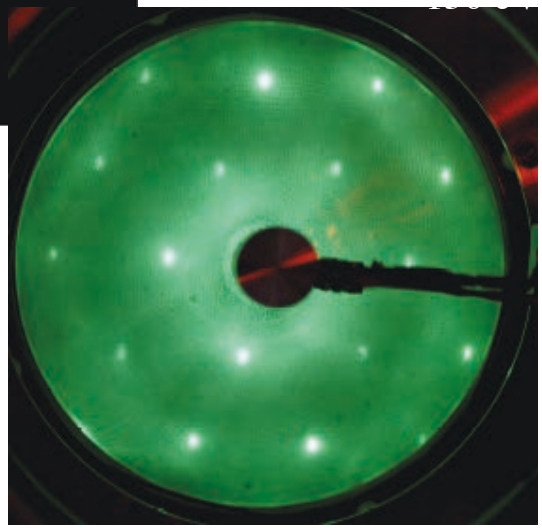


Real Space

a = 2.854 Å

b = 5.869 Å

c = 4.955 Å

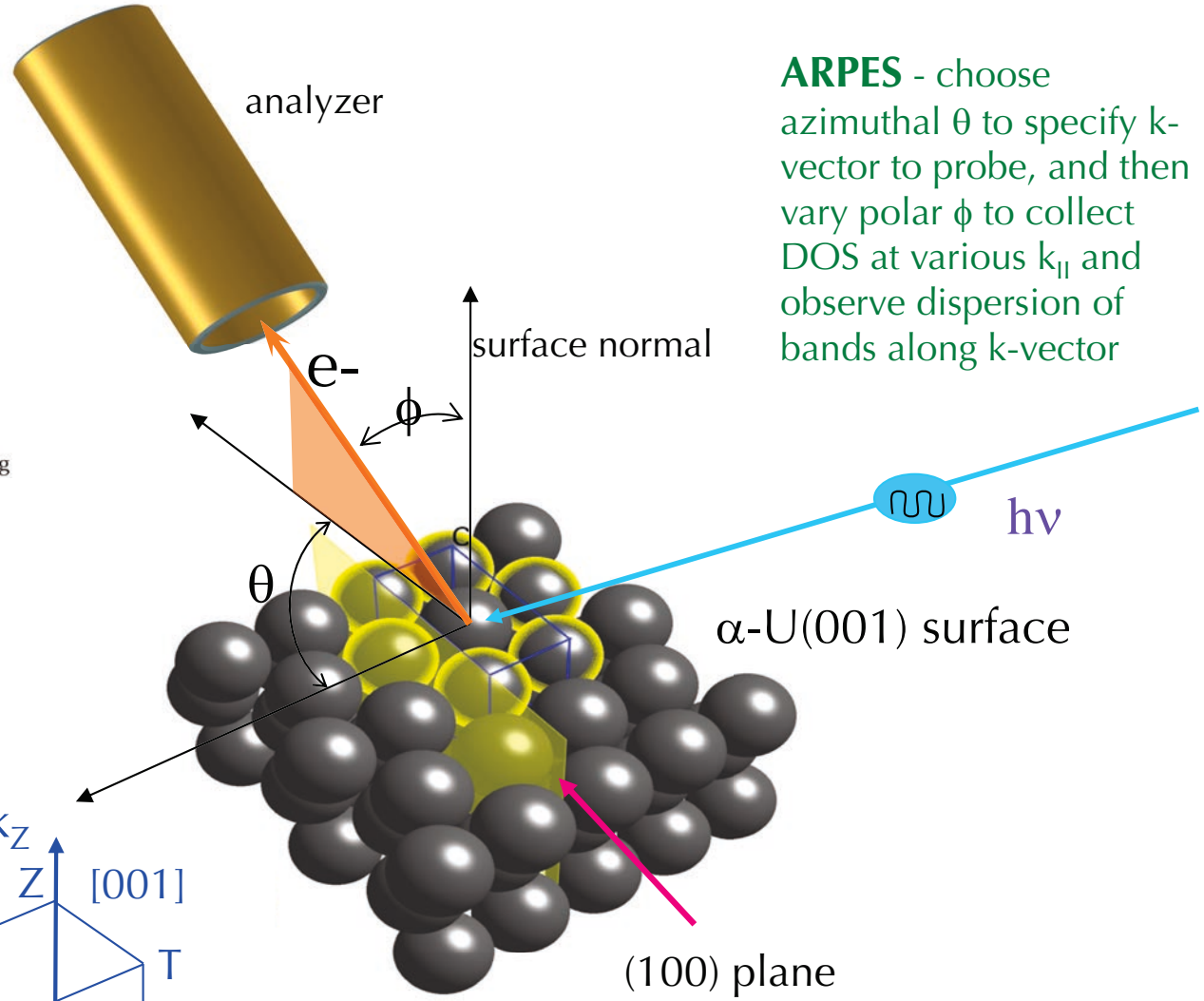


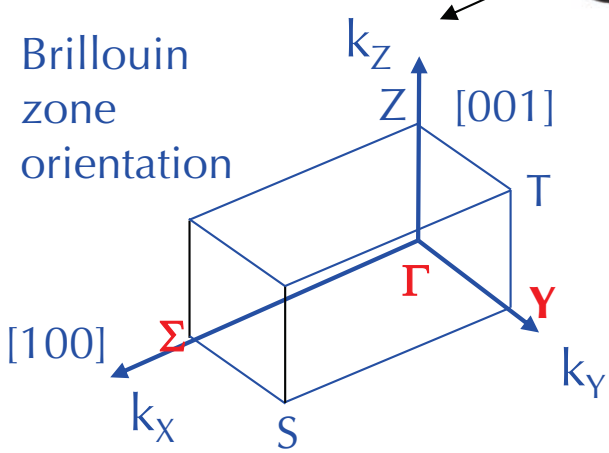
- First U(001) LEED pattern, long range order bulk termination
- Calculation & Experiment < 2 % difference
- Structural reactivity studies: O_2, D_2, H_2
- I-V analysis underway

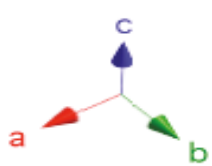
ARPES - choose azimuthal θ to specify k-vector to probe, and then vary polar ϕ to collect DOS at various $k_{||}$ and observe dispersion of bands along k-vector

$$E_{kinetic} = h\nu - e\Phi - E_{binding}$$

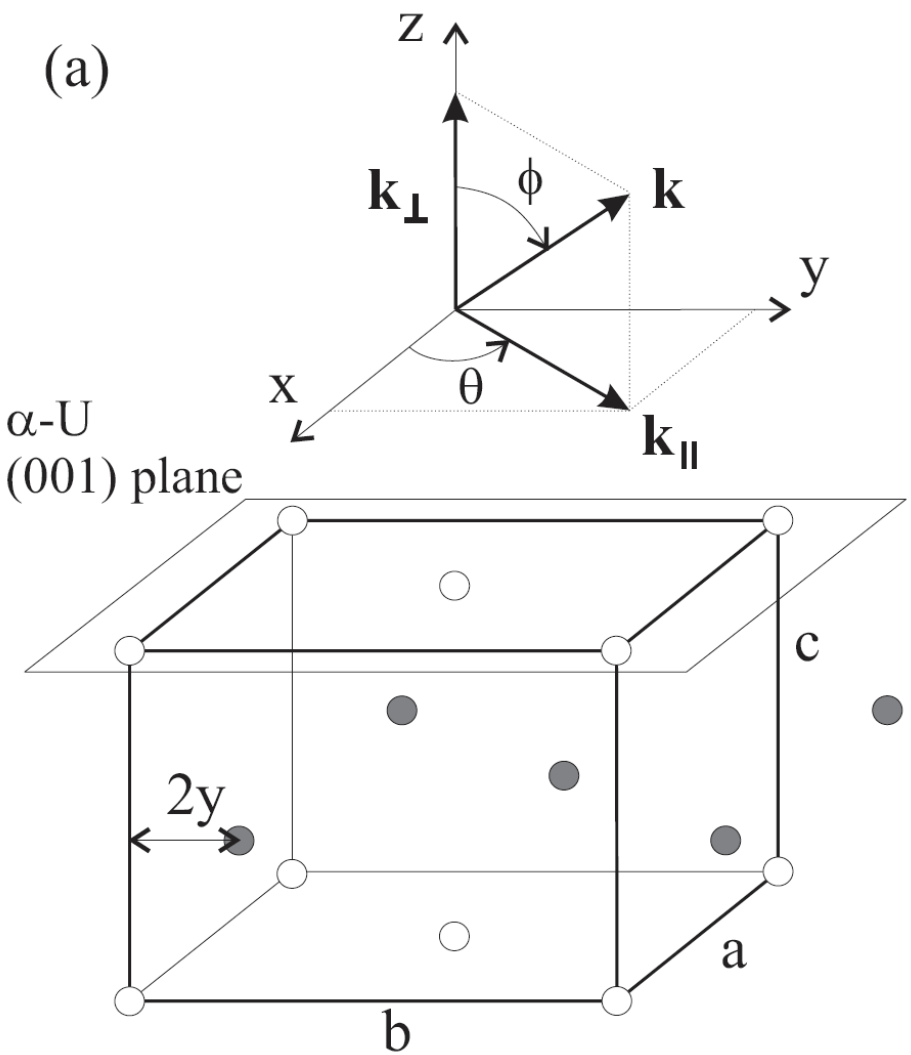
$$k_{||} = \sqrt{\frac{2m_e E_{kinetic}}{\hbar^2}} \cdot \sin\phi$$



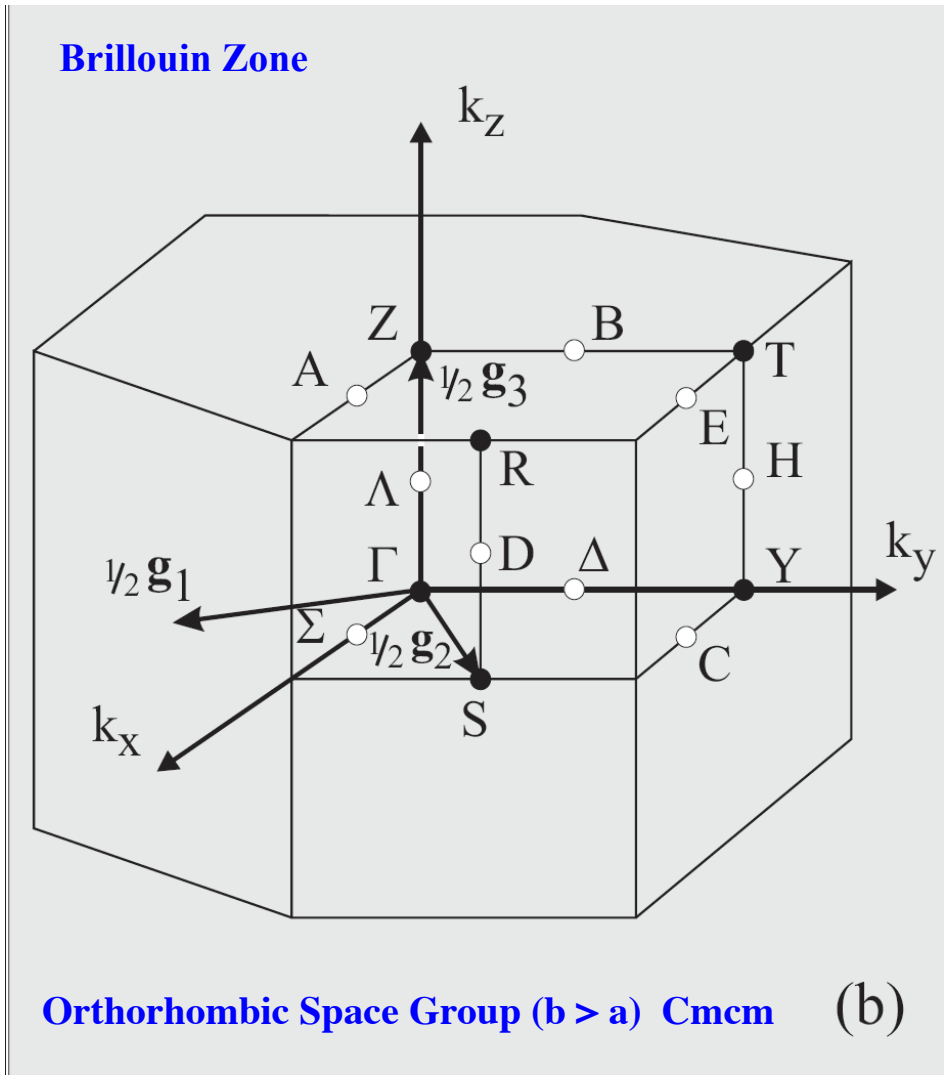




α - Uranium

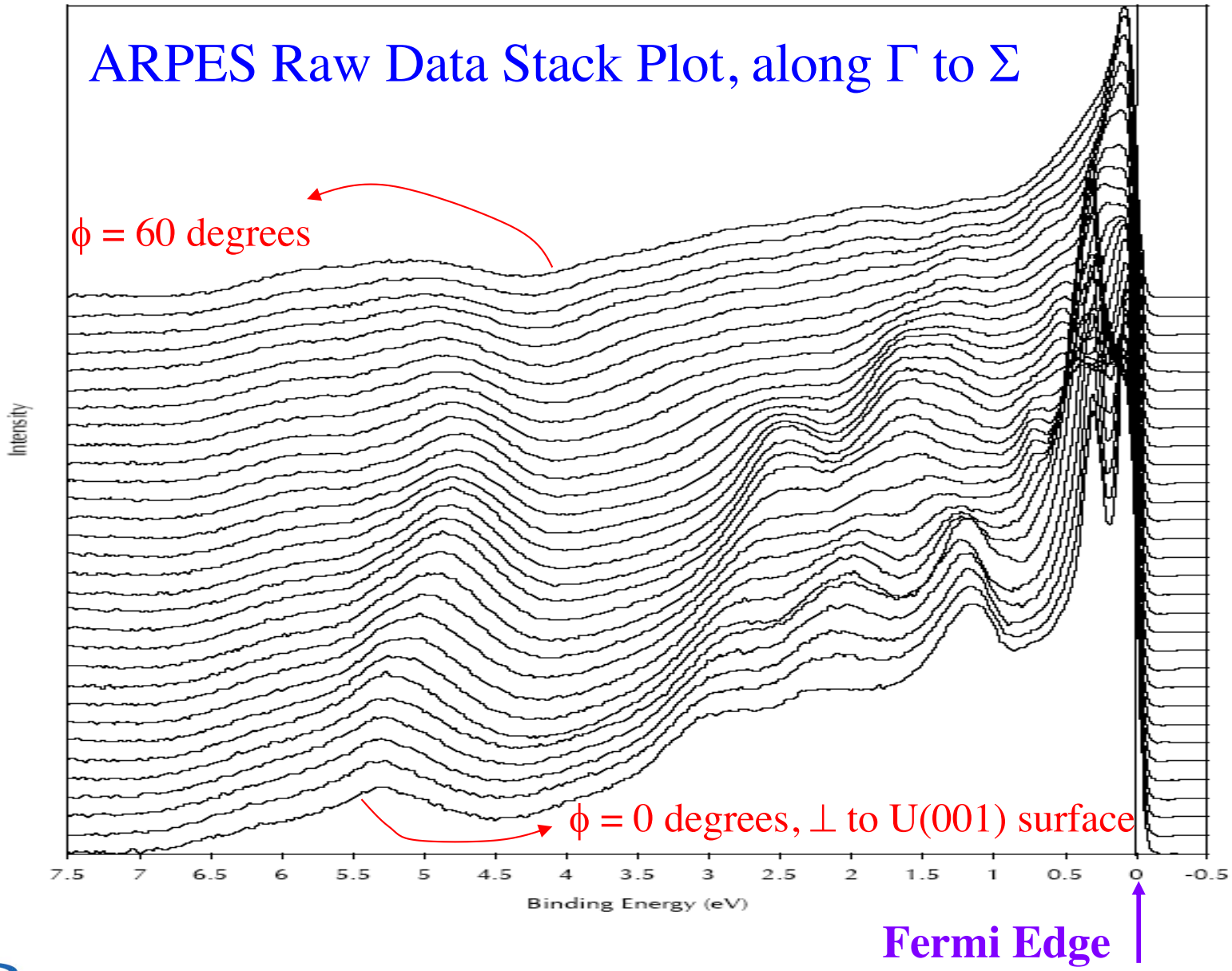


unit cell

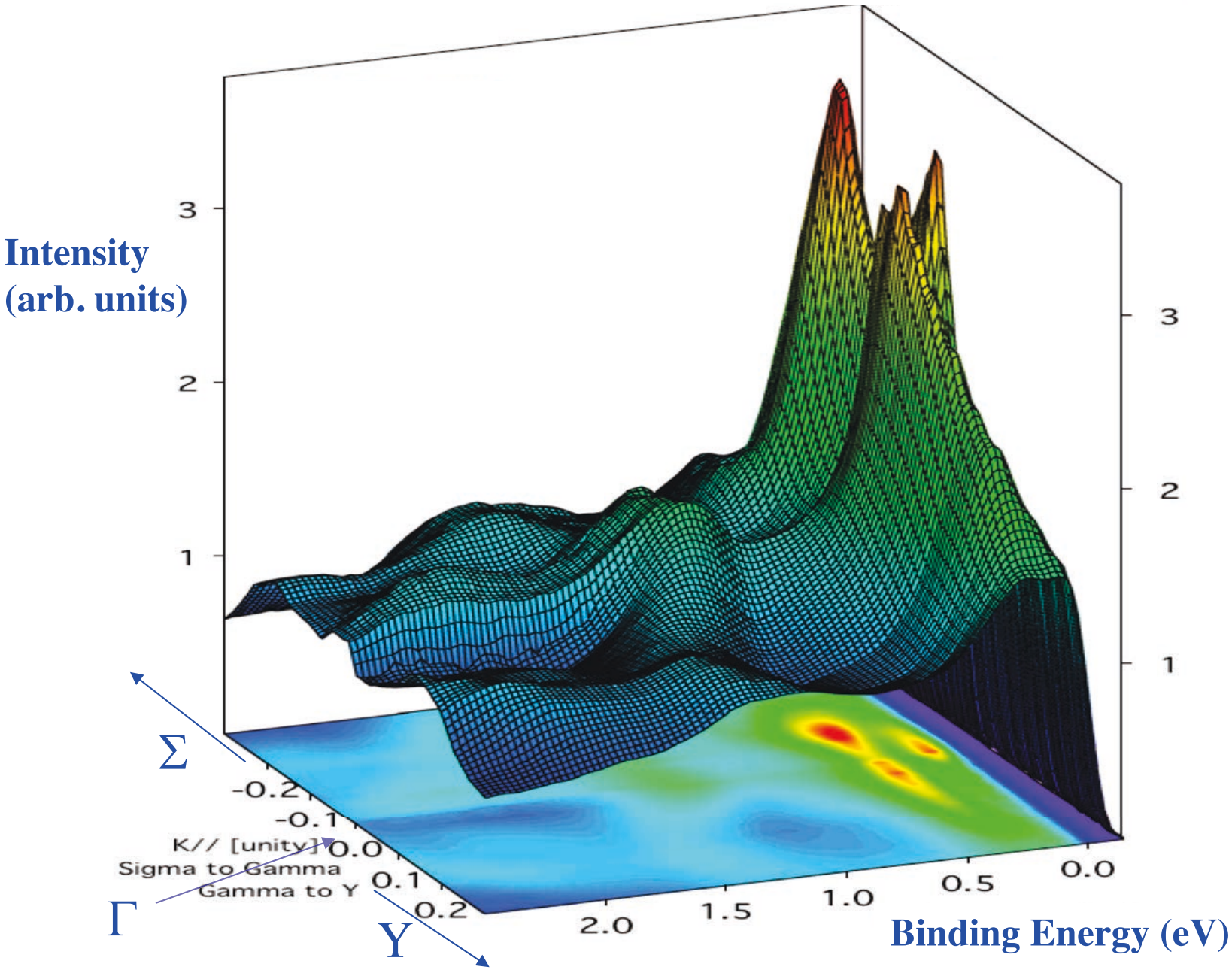


T. Hahn, *International Tables for Crystallography*, Dordrecht: The Netherlands, Kluwer Academic, 1992.

ARPES Raw Data Stack Plot, along Γ to Σ

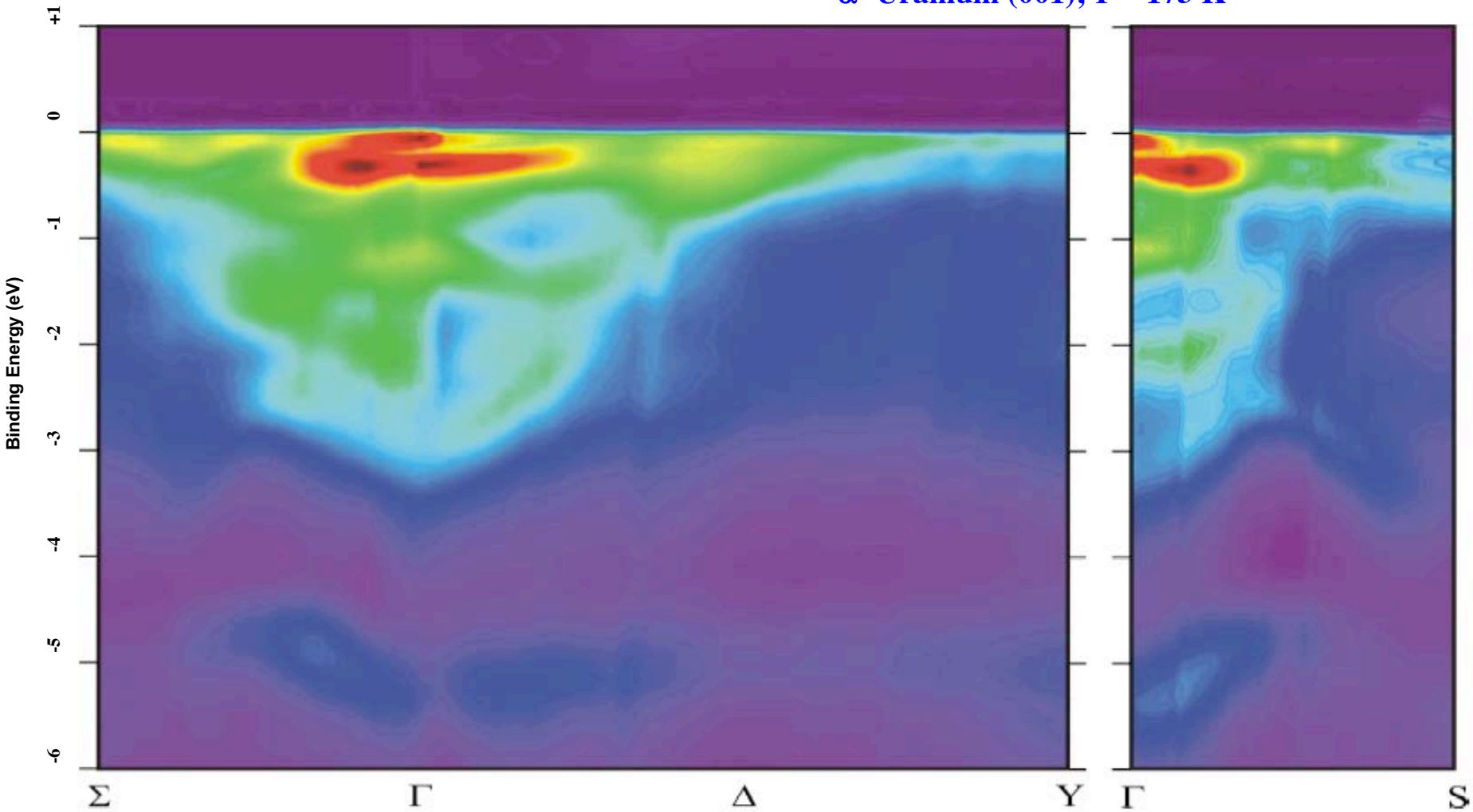


Uranium Band Map Sigma to Gamma, Gamma to Y, T = 173 K



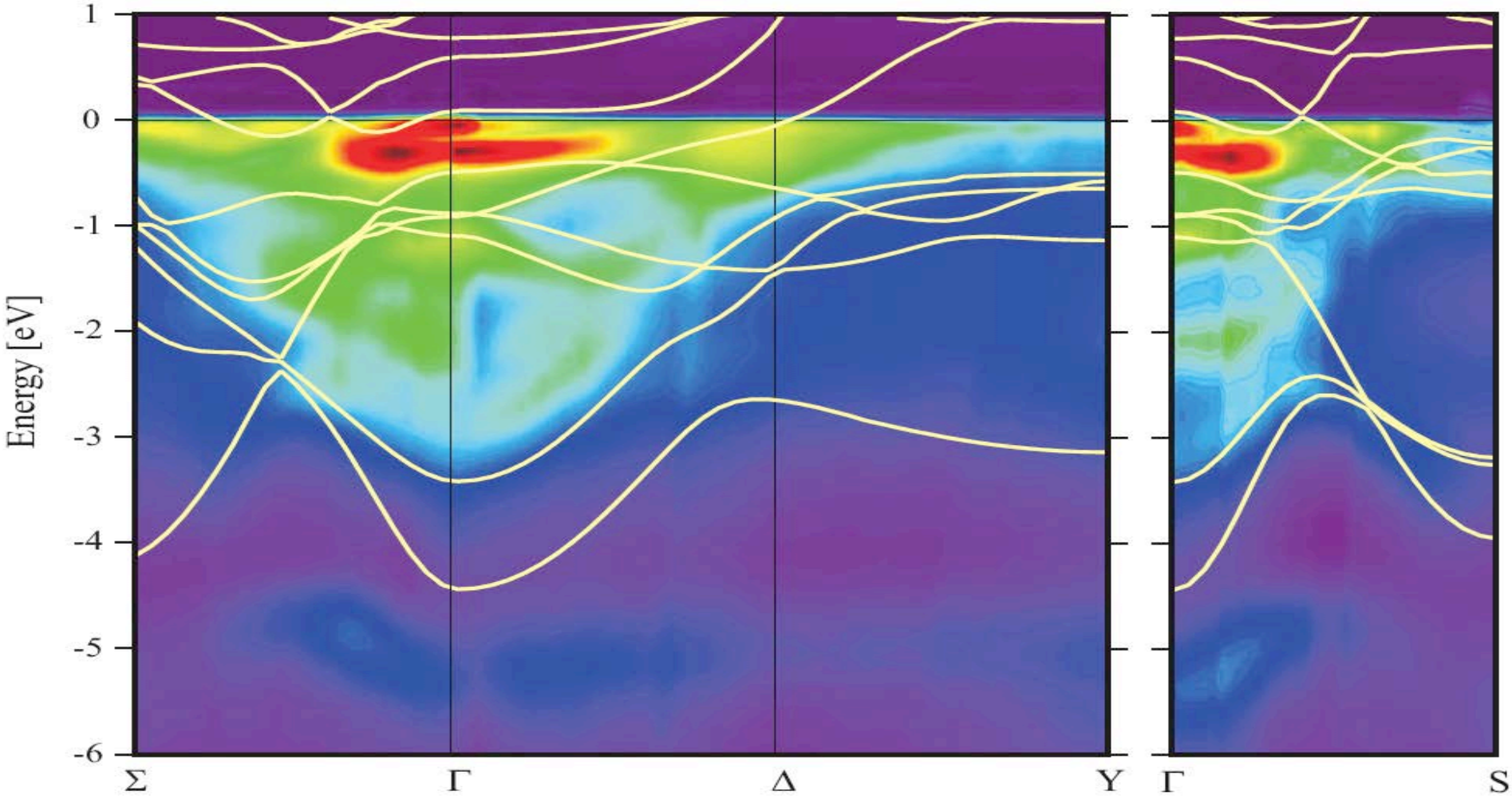
ARPES: Σ - Γ - Y, Γ - S

α -Uranium (001), T = 173 K

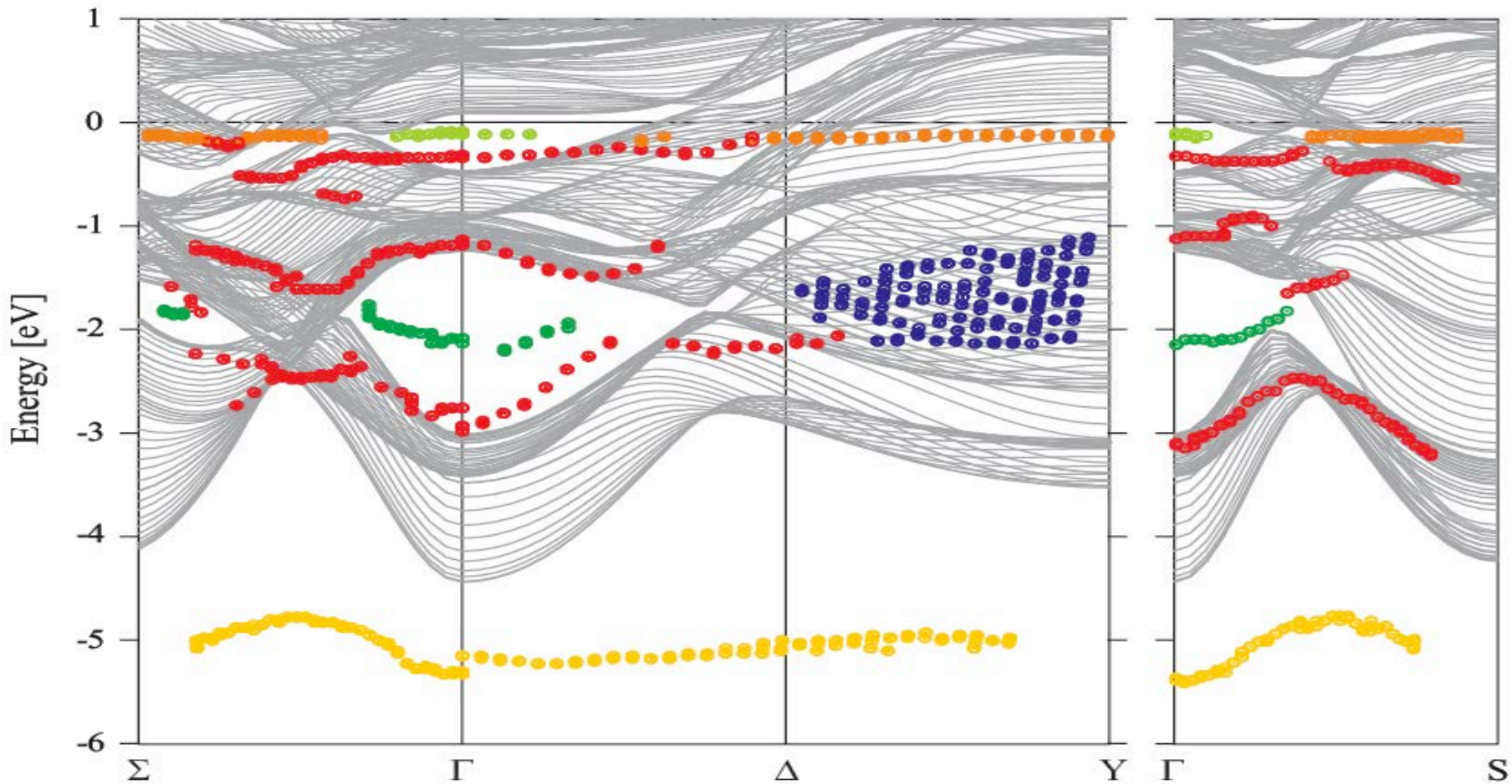


ARPES & WIEN2K

α -Uranium (001), T = 173 K



Data Maxima, DDOS WIEN2K Calculations



Band Structure

Band Structure, Low Intensity

d-Band, Tamm surface states

f-Band, f-like, Shockley surface states

Plasmon ??

Flat band near FE due to disorder ?

Conclusions:

- Single crystal U(001) samples yield expected experimental LEED pattern (< 2% difference). Showing long range bulk termination.

- ARPES data exhibit band dispersion at 173 K and correlates favorably with DDOS WIEN2K, if temperature and e^- correlations are considered.

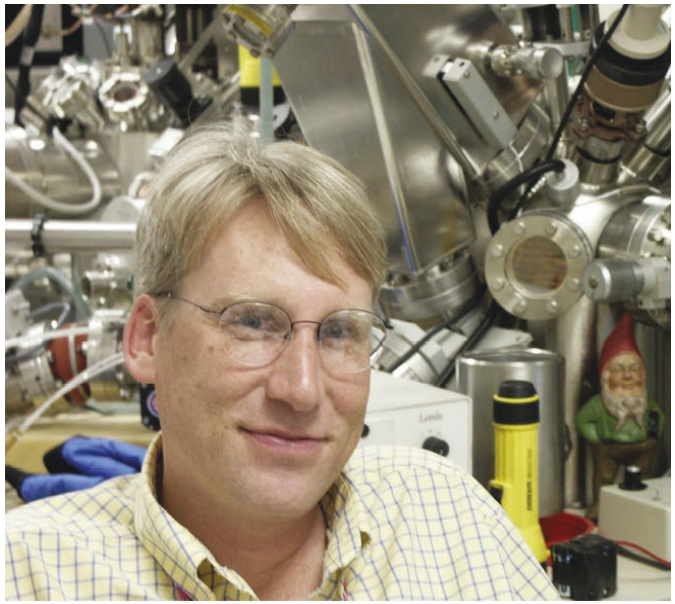
Next Steps:

- Colder Temperatures, Higher Resolution

- Fermi Surface Mapping, at ALS, Berkeley, CA

With special thanks to my advisors:

J. L. Smith



Roland K. Schulze

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Funded by: Department of Energy

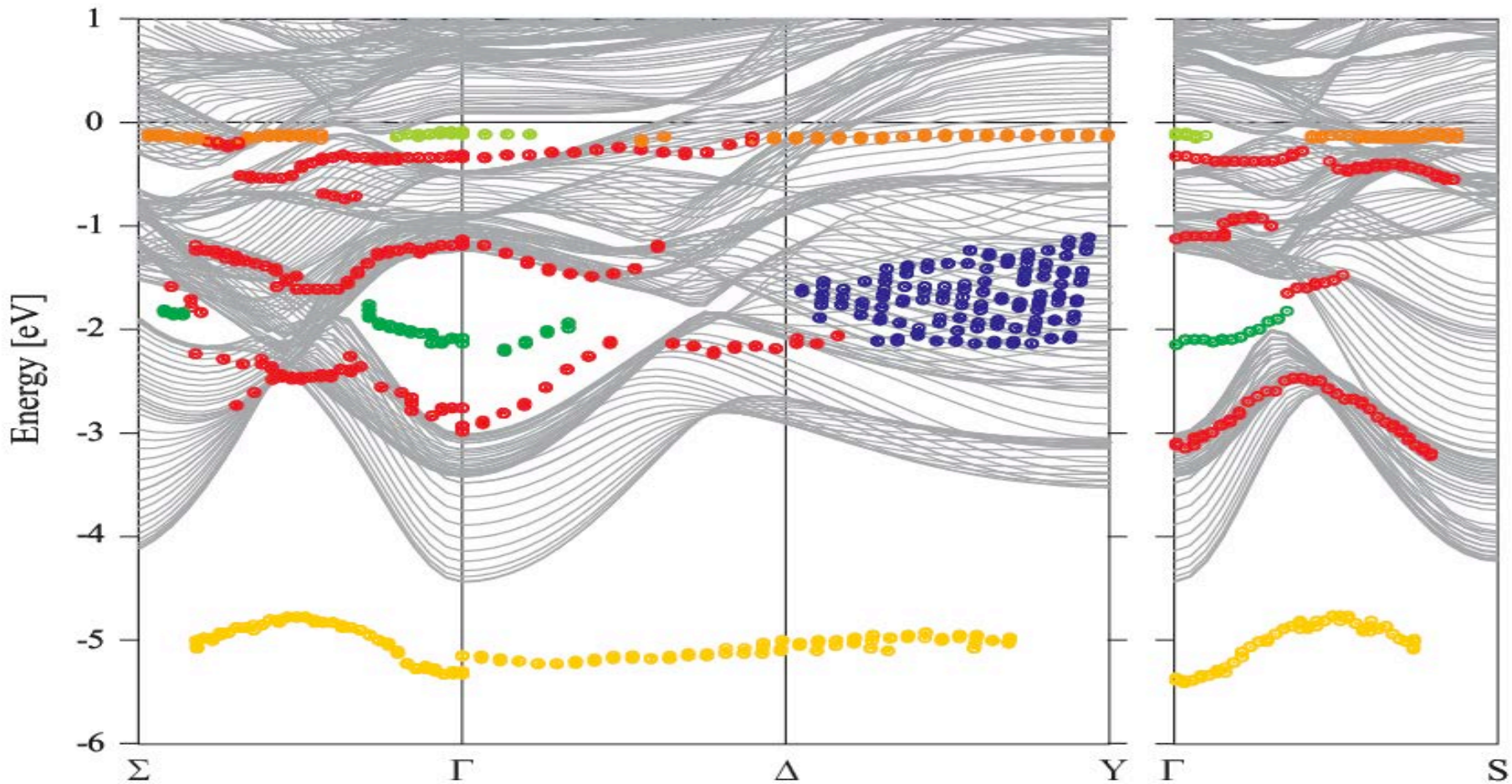
Conclusions:

- Single crystal U(001) samples yield expected experimental LEED pattern (< 2% difference). Showing long range bulk termination.
- ARPES data exhibit band dispersion at 173 K and correlates favorably with DDOS WIEN2K.
- Narrow f-band near Fermi edge. Presence of d and f surface states.
- Data-Theory mismatch possibly due to: ΔT, e^- to e^-, e^- to phonon contributions.

Next Steps:

- Colder Temperatures, Higher Resolution
- Fermi Surface Mapping, at ALS, Berkeley, CA
- Matrix Element Calculation, DMFT Calculation

Data Maxima, DDOS WIEN2K Calculations



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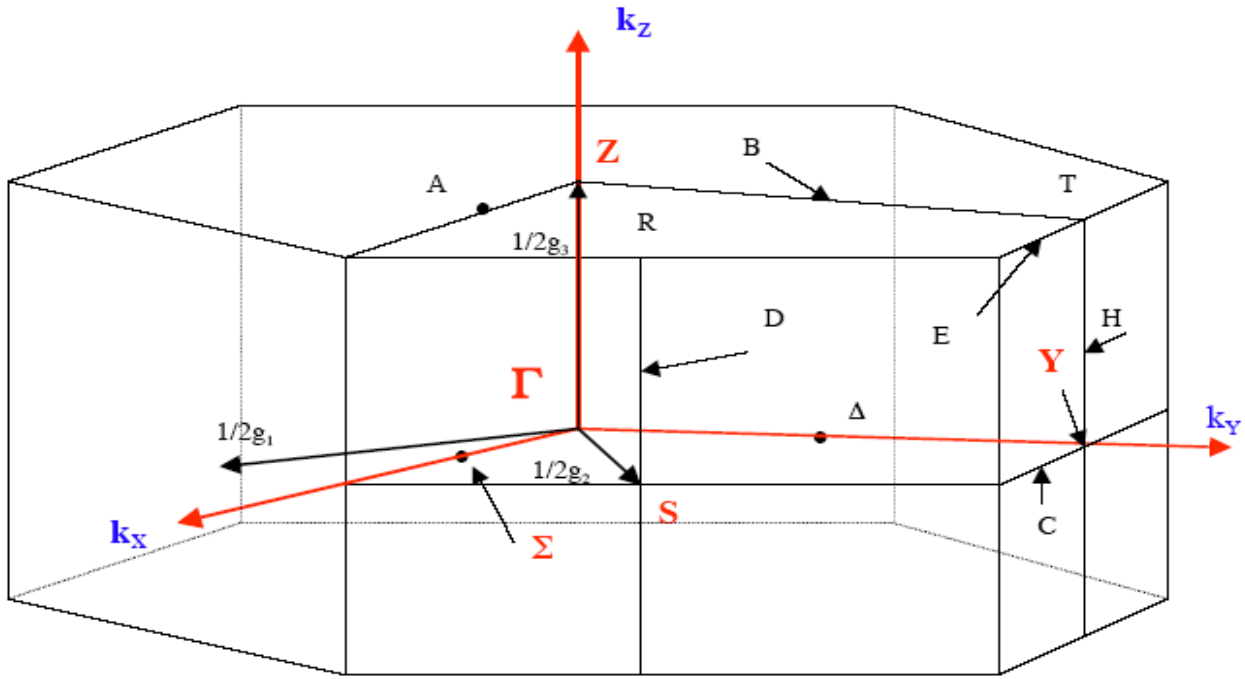
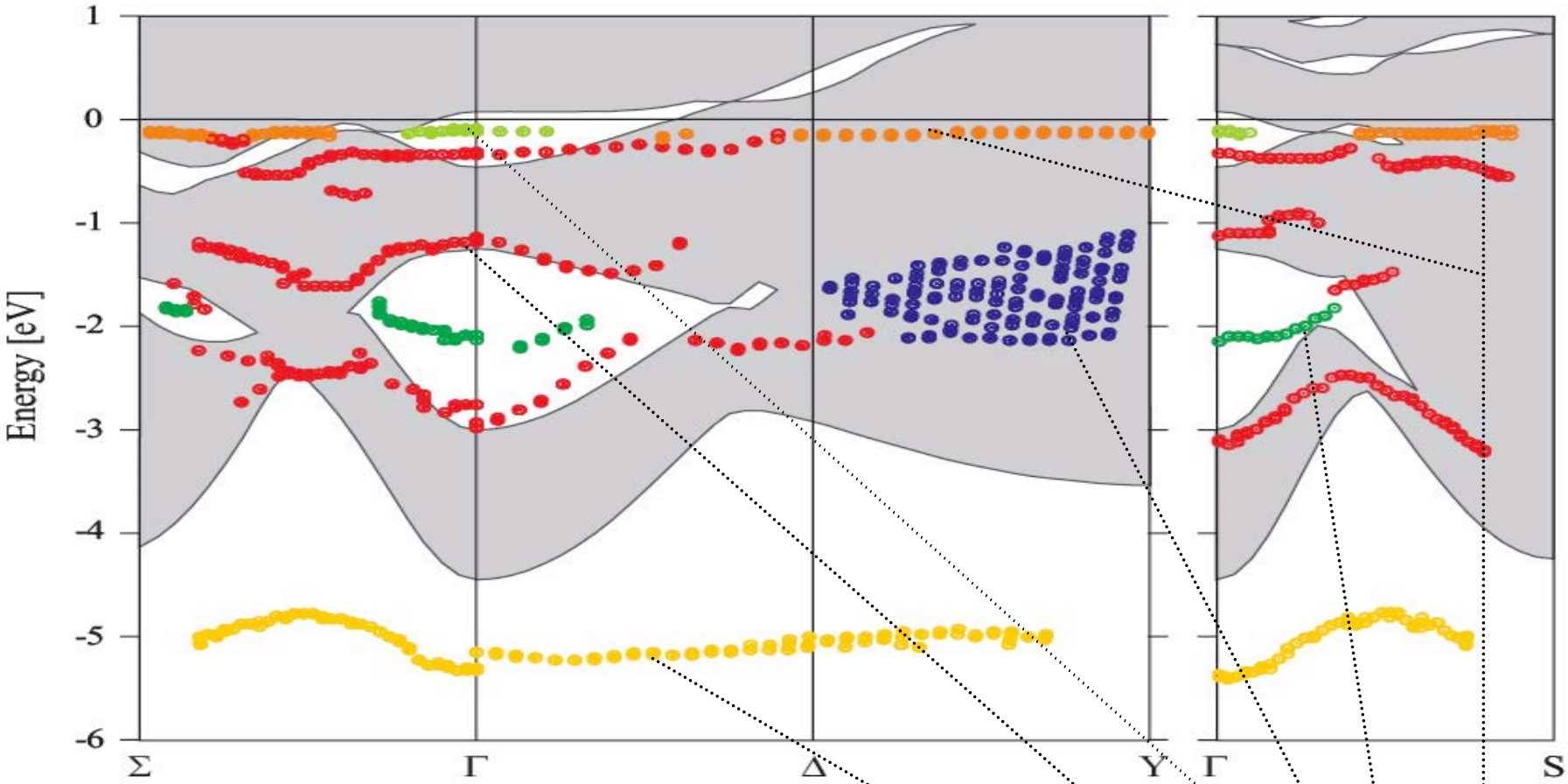
Brillouin Zone



Orthorhombic Space Group, b > a, α - Uranium

T. Hahn, *International Tables for Crystallography*,
Dordrecht: The Netherlands, Kluwer Academic, 1992.

Data Maxima, WIEN2K Gap Calculations & Surface States



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